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POSTER

OPT-AIL: Provably & Practically Efficient Adversarial Imitation Learning with General Function Approximation

The first provably efficient online AIL under general function approximation — and it needs only two simple objectives.

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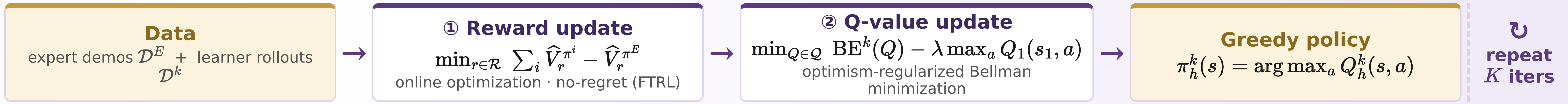
Paper & Code

OPT-AIL LOOP

Each iteration solves just **two optimization problems** — easy to implement with deep nets.



Data
expert demos $\mathcal{D}^E$ + learner rollouts $\mathcal{D}^k$
→
① Reward update
$\min_{r\in\mathcal{R}} \sum_i \widehat{V}_r^{\pi^i} - \widehat{V}_r^{\pi^E}$
online optimization · no-regret (FTRL)
→
② Q-value update
$\min_{Q\in\mathcal{Q}} \mathrm{BE}^k(Q) - \lambda \max_a Q_1(s_1,a)$
optimism-regularized Bellman minimization
→
Greedy policy
$\pi_h^k(s) = \arg\max_a Q_h^k(s,a)$
↻ repeat K iters

1 The Theory–Practice Gap

Adversarial imitation learning (AIL) drives state-of-the-art imitation with neural networks and often outperforms behavioral cloning (BC) — yet its theory lags far behind practice.

- Guarantees exist only for **tabular/linear** settings — not the general function approximation used in practice.
- Provable methods need **count-** or **covariance-matrix bonuses** that are hard to implement with neural networks.

Q: Can AIL be *provably efficient* under general function approximation *and* simple to implement?

2 Online AIL Setup

AIL imitates the expert through an adversarial game: recover a reward that maximizes the value gap, then a policy that minimizes it under that reward.

ADVERSARIAL OBJECTIVE

$$\min_{\pi} \max_{r} V_r^{\pi^E} - V_r^{\pi}$$

Aim: a small **imitation gap** $V^{\pi^E} - V^{\pi}$ from few **expert trajectories** (expert sample complexity) and few **environment interactions** (interaction complexity).

General function approximation: a reward class $\mathcal{R}$ and Q-value class $\mathcal{Q}$, under standard **realizability** (of $\mathcal{R}$ and $\mathcal{Q}$) and **Bellman completeness** of $\mathcal{Q}$.

3 Contributions

- **First provably efficient** online AIL with **general function approximation**.
- Polynomial expert-sample & interaction complexity; provably beats **BC** by $\mathcal{O}(H^2)$ in expert samples.
- Reduces to just **two optimization objectives** → a direct neural-network implementation.
- Matches or beats prior deep AIL methods (**IQLearn, PPIL, FILTER, HyPE**) and BC on **8 DMControl** tasks.

4 OPT-AIL: Two Objectives

Imitation gap = **reward error** + **policy error**; control each:

① REWARD ERROR → ONLINE OPTIMIZATION

$$\mathcal{L}^i(r) = \widehat{V}_r^{\pi^i} - \widehat{V}_r^{\pi^E}, \quad r^k = \text{no-regret}\big(\{\mathcal{L}^i\}_{i<k}\big)$$

Off-policy reward learning, justified via FTRL.

② POLICY ERROR → OPTIMISM-REGULARIZED BELLMAN MIN

$$\min_{Q\in\mathcal{Q}} \mathrm{BE}^k(Q) - \lambda \max_a Q_1(s_1,a)$$

Optimism drives exploration while the Bellman error keeps Q accurate; the policy is greedy(Q^k).

5 Theoretical Guarantee ★ main result

Under realizability, Bellman completeness, and a low **generalized eluder coefficient** d_{GEC}, OPT-AIL learns an ε-optimal policy (w.h.p.) with:

Theorem.
Expert samples $N = \widetilde{\mathcal{O}}\big(H^2 \log \mathcal{N}(\mathcal{R})/\varepsilon^2\big)$
Interactions $K = \widetilde{\mathcal{O}}\big((H^4 d_{\mathrm{GEC}} \log(\mathcal{N}(\mathcal{Q})\mathcal{N}(\mathcal{R})) + H^2)/\varepsilon^2\big)$

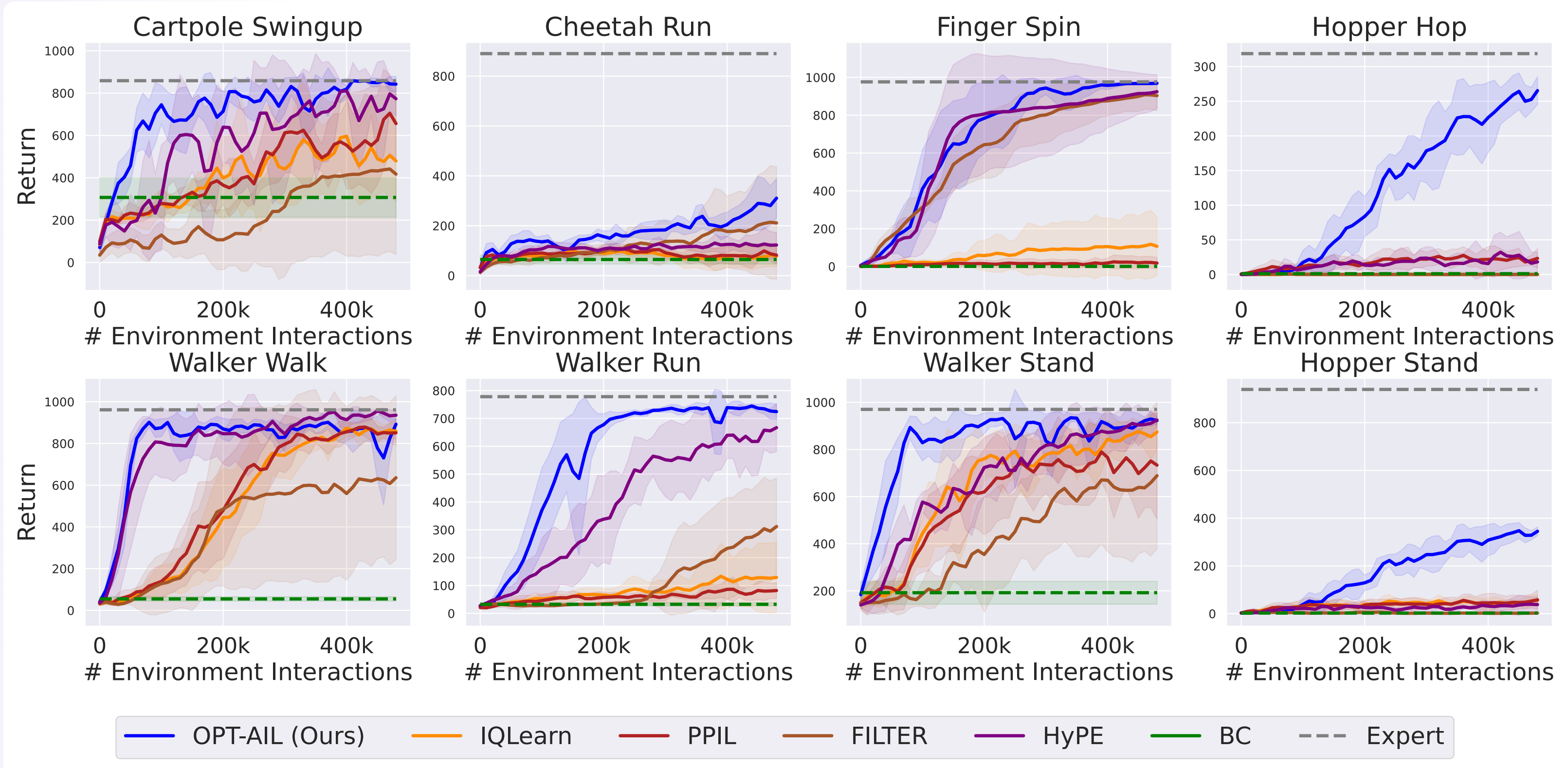
d_{GEC} (Zhong et al., 2022) covers tabular, linear & low Bellman-eluder MDPs. The $\mathcal{O}(H^2)$ gain over BC provably **curbs compounding errors**.

6 Complexity vs. Prior Work

Setting	Algorithm	Expert samples	Interactions
Gen. func.	**BC**	$\widetilde{\mathcal{O}}(H^4 \log \lvert\Pi\rvert/\varepsilon^2)$	0
Tabular	**MB-TAIL**	$\widetilde{\mathcal{O}}(H^{3/2}\lvert\mathcal{S}\rvert/\varepsilon)$	$\widetilde{\mathcal{O}}(H^3\lvert\mathcal{S}\rvert^2\lvert\mathcal{A}\rvert/\varepsilon^2)$
Linear	**BRIG**	$\widetilde{\mathcal{O}}(H^2 d/\varepsilon^2)$	$\widetilde{\mathcal{O}}(H^4 d^3/\varepsilon^2)$
Gen. func.	**OPT-AIL**	$\widetilde{\mathcal{O}}(H^2 \log \mathcal{N}(\mathcal{R})/\varepsilon^2)$	$\widetilde{\mathcal{O}}(H^4 d_{\mathrm{GEC}} \log(\mathcal{N}(\mathcal{Q})\mathcal{N}(\mathcal{R}))/\varepsilon^2)$

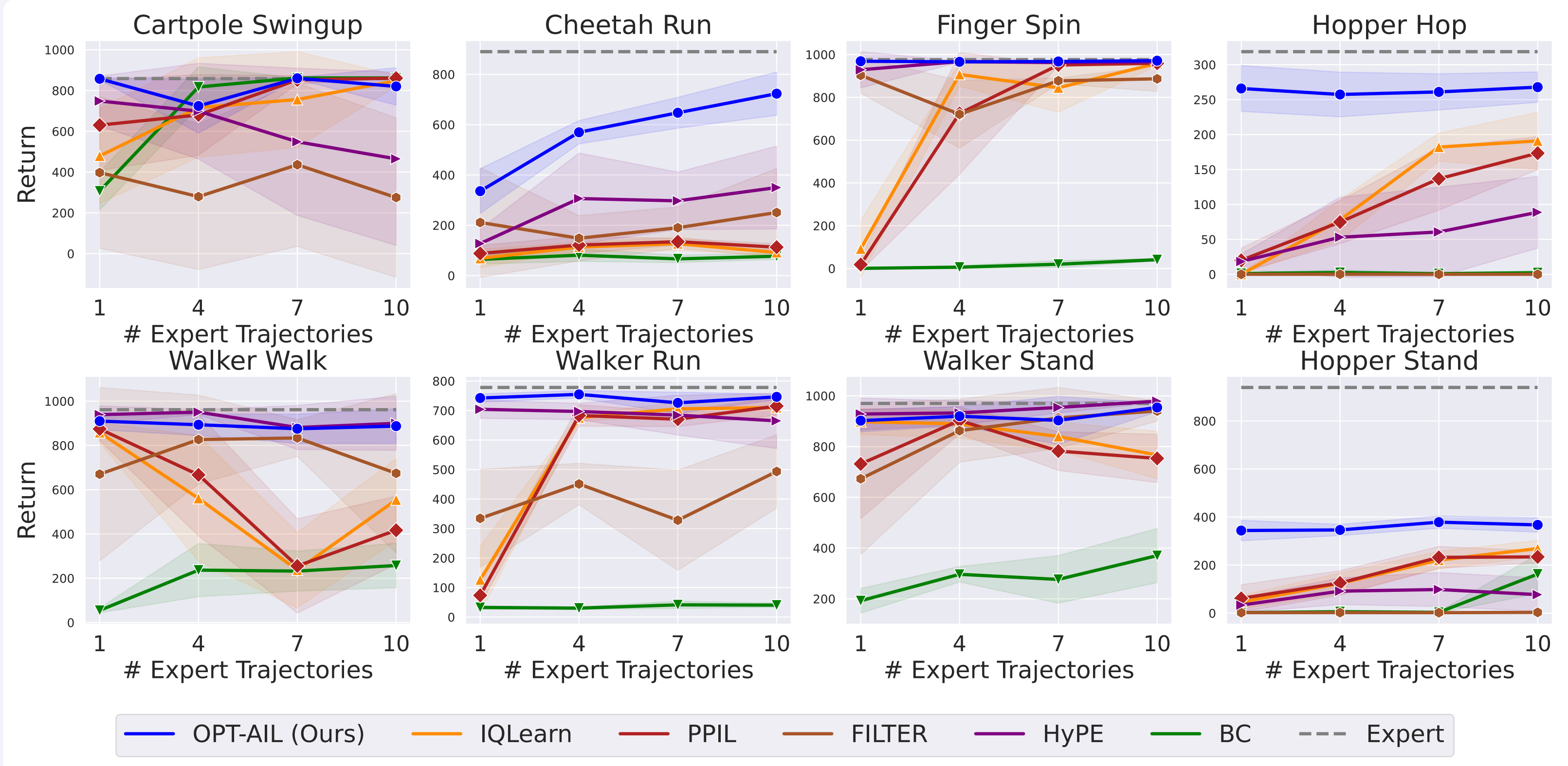
Prior provable AIL is **tabular/linear** only; OPT-AIL is the first under **general function approximation**.

7 Interaction Efficiency (1 expert trajectory) ★ headline



Learning curves · 8 DMControl tasks · 5 seeds. **OPT-AIL (blue)** matches or beats every SOTA deep AIL baseline on **all 8 tasks**, reaching **near-expert with far fewer interactions** on Cheetah Run, Finger Spin & Hopper Hop.

8 Expert-Sample Efficiency — the low-data regime



Final return vs. # expert trajectories (500k interactions). **OPT-AIL matches or exceeds** prior SOTA at every budget; with **just 1 expert trajectory** it is uniquely (near-)expert on Finger Spin, Walker Run & Hopper Hop, while **BC** (green) trails.

9 Takeaways

IDEA
First **provably efficient** AIL under general function approximation.

METHOD
Just **two objectives**: online reward optimization + optimism-regularized Bellman min.

THEORY
$\widetilde{\mathcal{O}}(H^2/\varepsilon^2)$ expert samples — $\mathcal{O}(H^2)$ **better** than BC.

PRACTICE
Matches or beats SOTA deep AIL on **8 DMControl** tasks, even with 1 trajectory.

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Paper: **arXiv:2411.00610** · Code: **github.com/LAMDA-RL/OPT-AIL**